

Domain Adaptation of Genomic Language Models for Plastic-Degrading Bacteria Discovery

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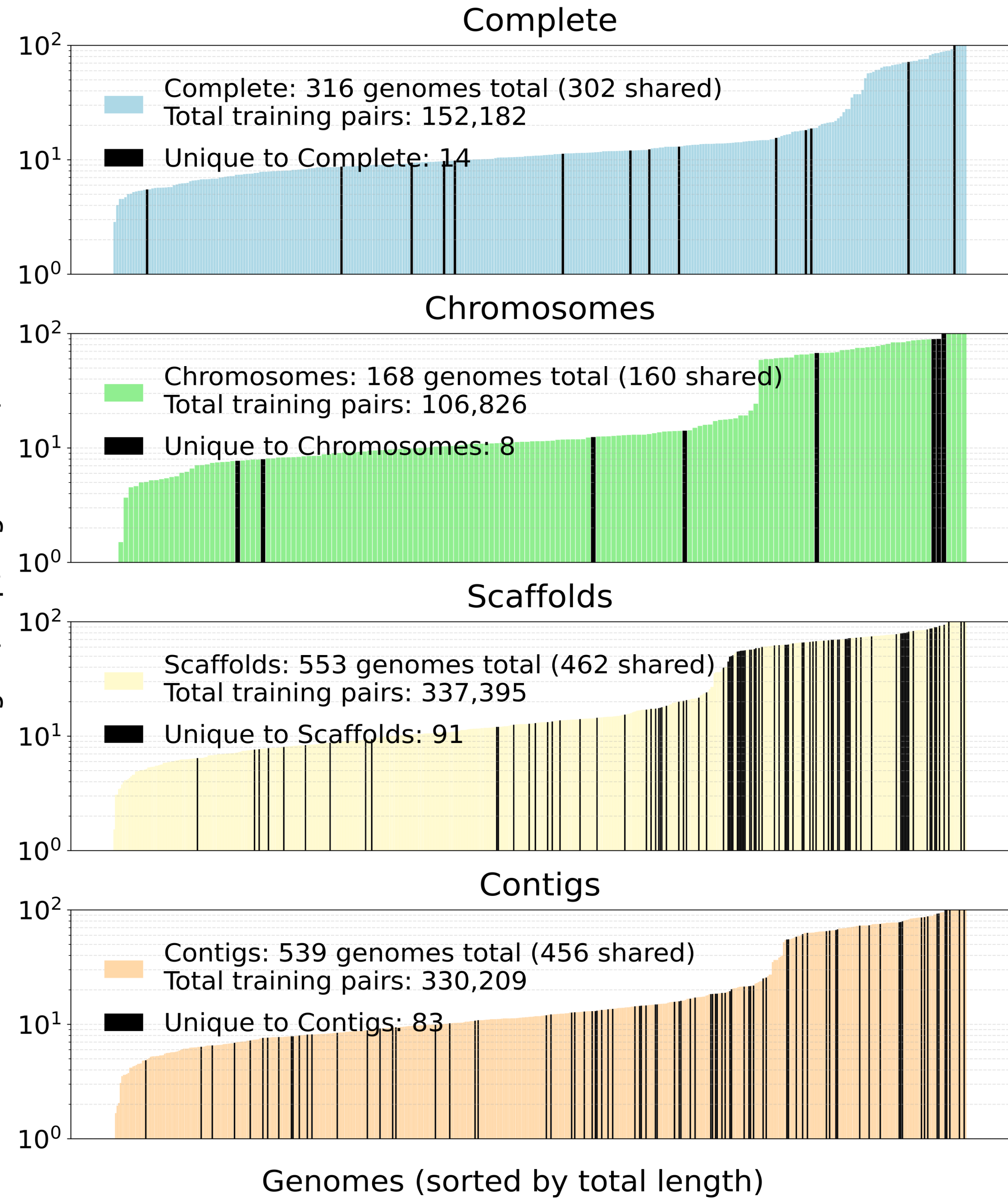
Introduction

Plastic pollution poses a growing environmental challenge and calls for scalable computational analysis of plastic-related microbial data, beginning as early as the sequence stage. Existing tools such as PEZy-Miner [1] and PlasticEnz [2] primarily target enzyme-level discovery.

In this work, we adapt two DNABERT [3,4] foundation models to known plastic-related DNA sequences and assess how fine-tuning reshapes genomic embedding space and clustering behavior. Fine-tuning is performed using plastic-related sequences derived from PlasticDB [5] genomes, chromosomes, scaffolds, and contigs. Our results provide insight into plastic-related signal at the genomic sequence level and help inform the development of plastic-aware genomic language models.

PlasticDB Data

Sequences from bacteria with known plastic-degrading capability were collected from PlasticDB across multiple assembly levels. Sequence length distributions for each are shown below:



References:

[1] Jiang et al. 2024. PEZy-miner: An AI-driven approach for the discovery of plastic-degrading enzyme candidates. *Metabolic Engineering Communications* 19, e00248.

[2] Krzynowek et al. 2026. PlasticEnz: An integrated database and screening tool combining homology and machine learning to identify plastic-degrading enzymes in meta-omics datasets. *PLoS Computational Biol* 22(1): e1013892.

[3] Zhihan et al. 2025. DNABERT-S: pioneering species differentiation with species-aware DNA embeddings, *Bioinformatics*, Volume 41, July 2025

[4] Zhihan et al. 2023. DNABERT-2: Efficient Foundation Model and Benchmark For Multi-Species Genome, arXiv:2306.15006, June 2023.

[5] PlasticDB - Plastic Biodegradation Database

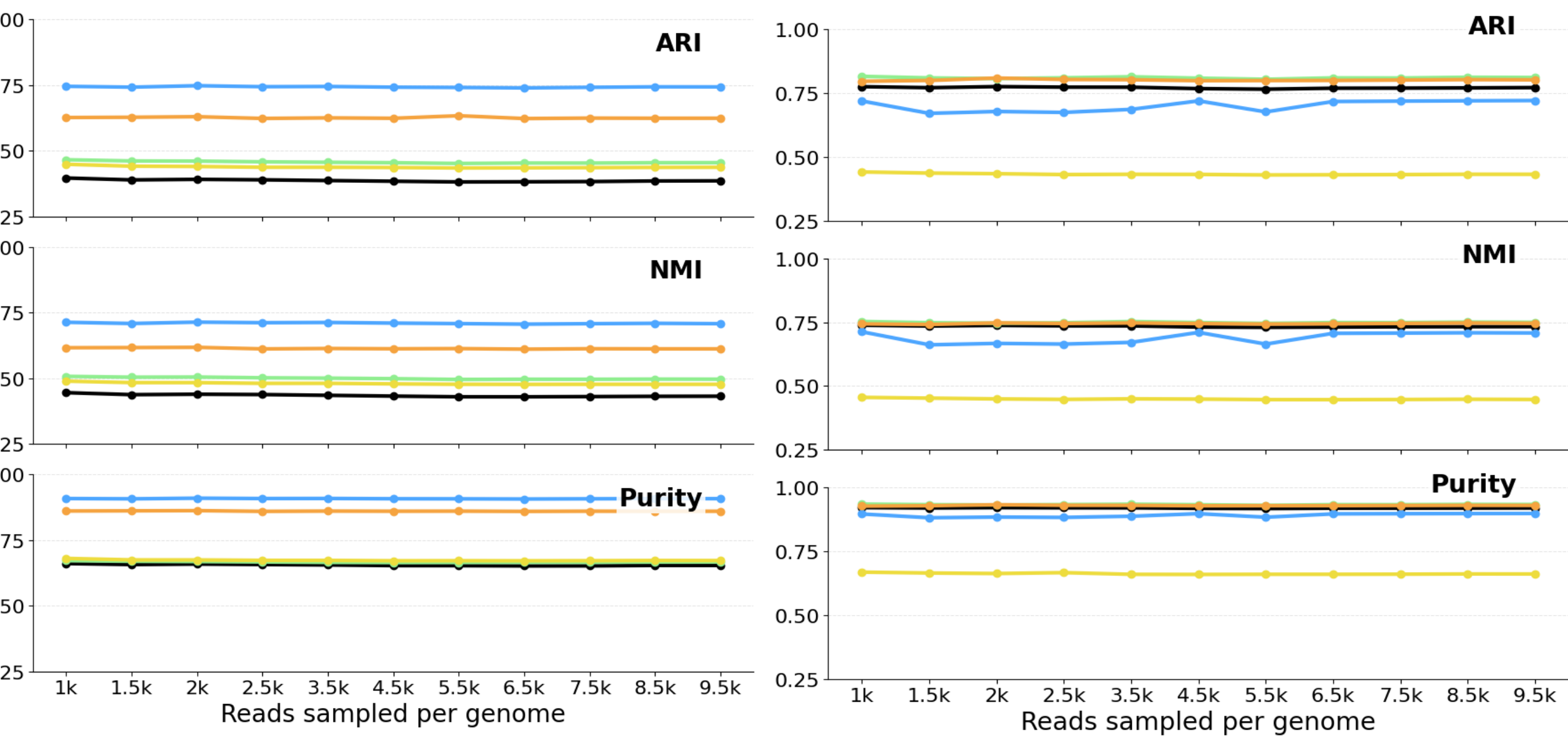
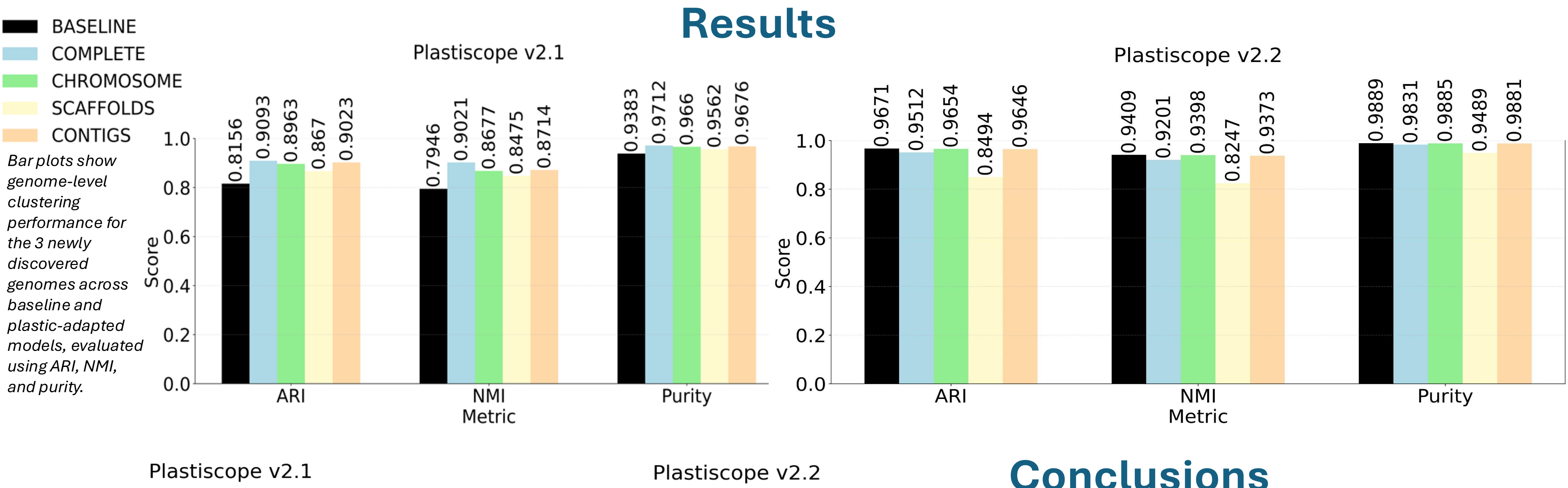
[6] Mitra and Stanescu, 2026, Complete genomic sequences of the single chromosome and four putative plasmids of *Sphingobium yanoikuyae* strain CC4533, *American Society for Microbiology Journals/Microbiology Resource Announcements* ASM/MRA

[7] Mitra and Stanescu, 2025, Complete genome sequence *Microbacterium* sp. strain Clip185, ASM/MRA

[8] Mitra and Stanescu, 2024, Complete genome sequence of *Acidovorax temperans* strain LMJ isolated from a contaminated *Chlamydomonas reinhardtii* Tris-Acetate-Phosphate medium culture plate, *American Society for Microbiology Journals/Microbiology Resource Announcements*

Methods

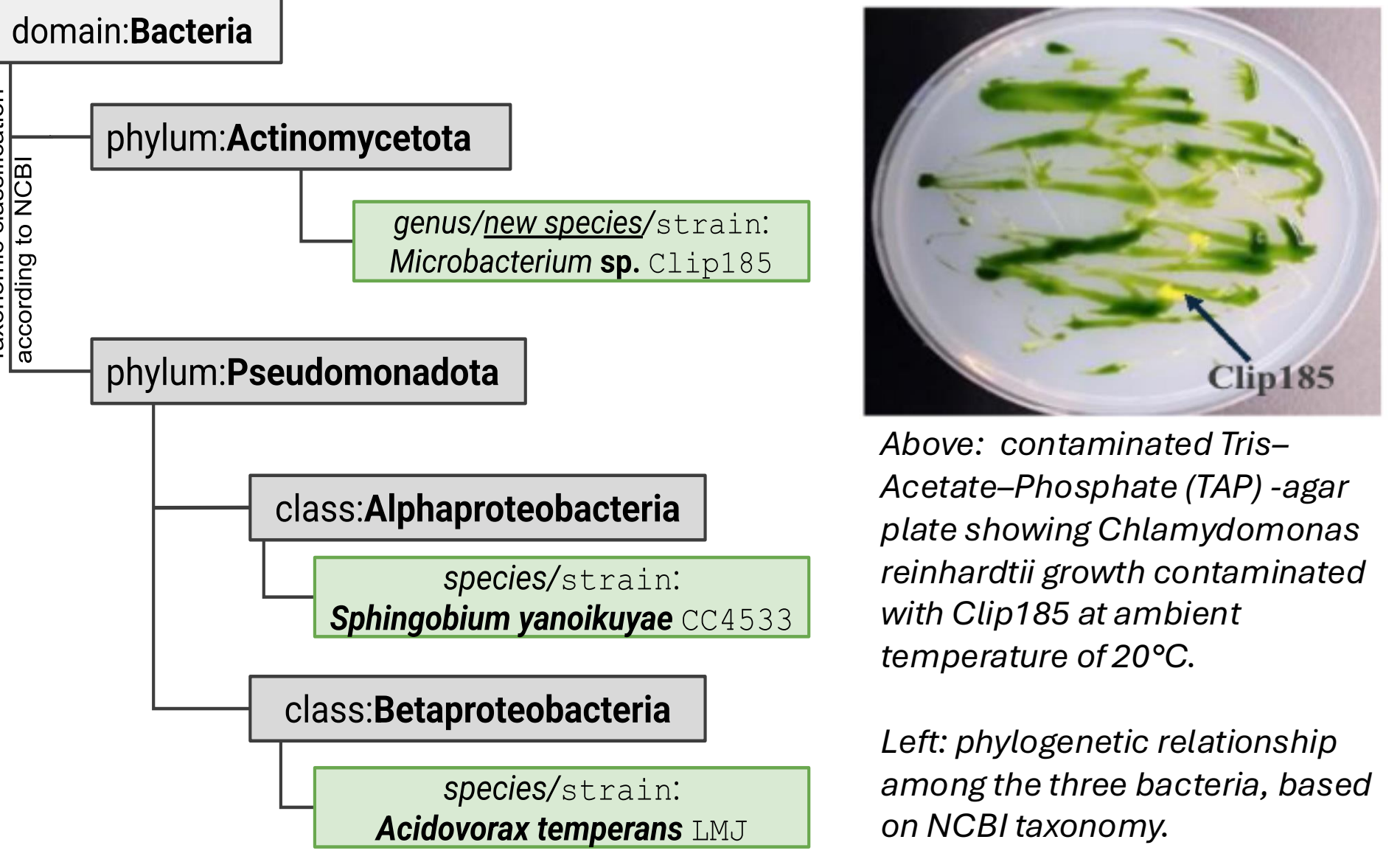
MODEL	DNABERT-2 (2024)	DNABERT-S (2025)	PLASTISCOPE	
			v2.1	v2.2
	General multi-species, transferable foundation models, 2.75B bp 135 species (incl. human, in/vertebrates, mammalian, fungi, protozoa, bacteria)			
Tokenizer	4096-token Byte Pair Encoding (BPE) vocabulary			
Parameters	117,068,544 trainable parameters			
Architecture	12-layer BertModel with 768 hidden dimensions MosaicBERT -style BertUnpadAttention and BertGatedLinearUnitMLP (GLU FFN), LayerNorm, and dropout ALiBi positional bias / no learned absolute positional embeddings			
Embeddings	4096 × 768 lookup table and input pipeline to encoder			
Training Objective/ Pretraining	- Pure Masked Language Modeling (MLM) - Randomly mask ~15% BPE	Start from	Start from	
		DNABERT-2	DNABERT-2	DNABERT-S
		weights and adds species-aware learning via - Simple Contrastive Learning (SimCLR) - Manifold Instance Mixup (MI-Mix) loss weights on +/- pairs to focus on harder-to-cluster sp. - Curriculum Contrastive Learning (C²LR)	weights and fine-tune to plastic via - an additional C²LR objective on - plastic-related sequence pairs for each dataset (contigs, scaffolds, chromosomes, genomes)	
Compute Setup	8 NVIDIA RTX 2080Ti GPU 11GB 14 days	8 NVIDIA A100 GPUs, FlashAttention 80GB ea. 2 days	1 NVIDIA TESLA T4 GPU 16GB 1 day for each full adaptation set	
Use cases	Designed for fine-tuning and downstream tasks.	Excels at metagenomics binning and is more versatile for taxonomic tasks.	Designed for downstream tasks for plastic-degrading bacterial signatures.	



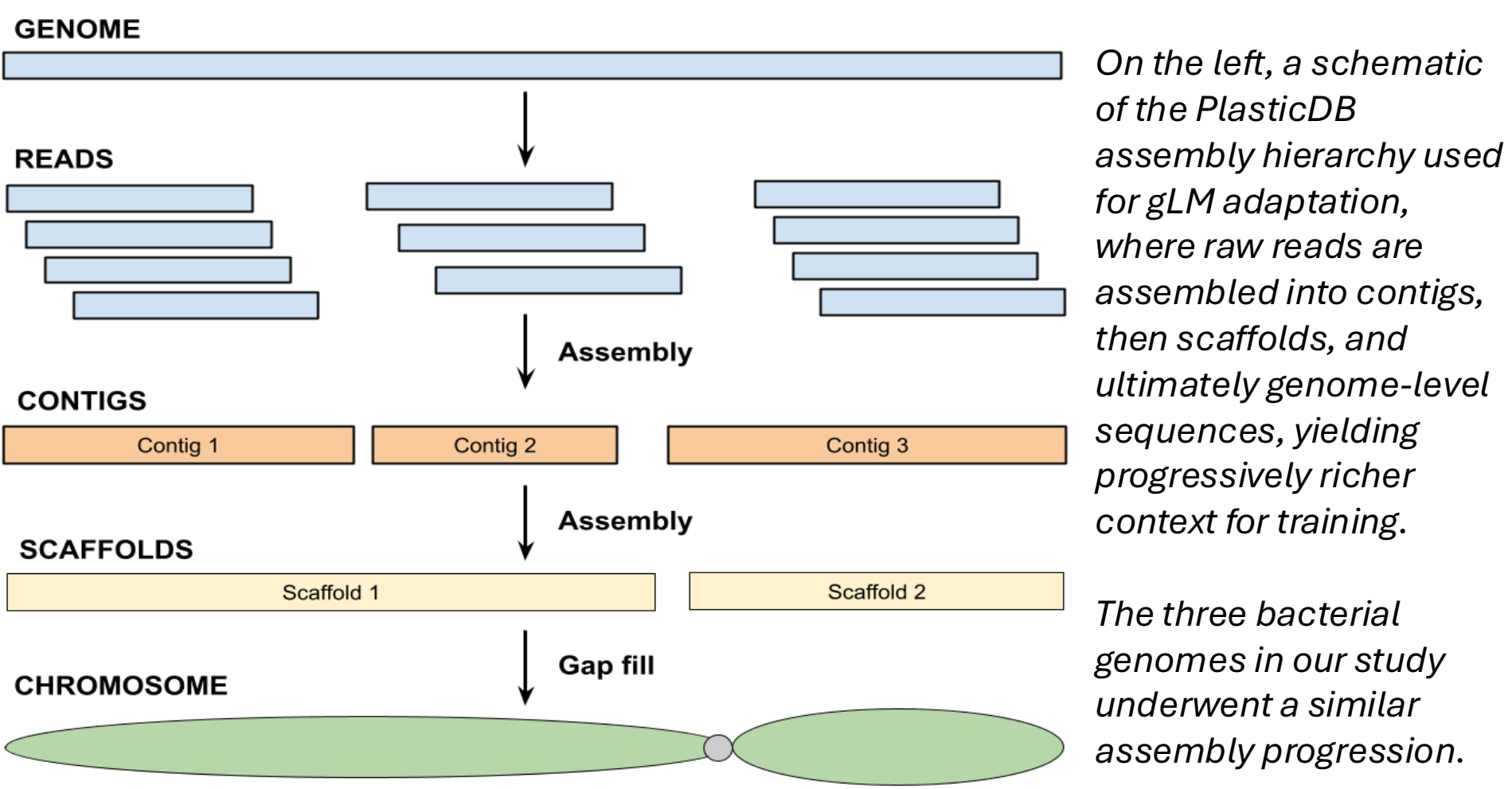
Read-level clustering performance across progressively larger nested subsets of reads from the three newly discovered genomes. Across both baseline and adapted models, ARI, NMI, and purity remain stable, with near-flat trajectories across scales, indicating that the embedding spaces remain well structured as the clustering problem grows.

Evaluation

The two baseline models and eight adapted models were evaluated on clustering three newly discovered sequenced bacteria [6, 7, 8] at the genome and read levels.



The three new bacterial genomes and their raw reads were partitioned into 2.5 kb sequence chunks, which were then tokenized and embedded into 768-dimensional vectors. K-means clustering was subsequently applied to the resulting chunk-level embeddings.



Conclusions

- At the **genome level**, both baseline and adapted models show strong clustering performance, indicating that the embedding spaces capture the dominant taxonomic structures. Cluster-level analysis further revealed that only Plastiscope v2.2 achieves perfect clustering of Clip185, the taxonomically most distant of the three organisms.
- At the **read level**, clustering remains stable under scaling for both model families. Adaptation benefits DNABERT-2 more than DNABERT-S: Plastiscope v2.1 shows the clearest gains over baseline, while v2.2 shows only modest gains and deteriorates under scaffold adaptation. In Plastiscope v2.1, purity exceeds ARI/NMI, indicating majority-homogeneous clusters but weaker agreement with the true partition.
- Superior metrics for the genome chunks indicate that longer, more coherent sequences improve species separability, whereas raw reads remain more challenging due to limited context, repeats, conserved regions, uneven coverage, and errors.
- These findings of our exploratory work support the feasibility of plastic-aware genomic adaptation and motivate further study of plastic-adapted models for downstream tasks.

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